

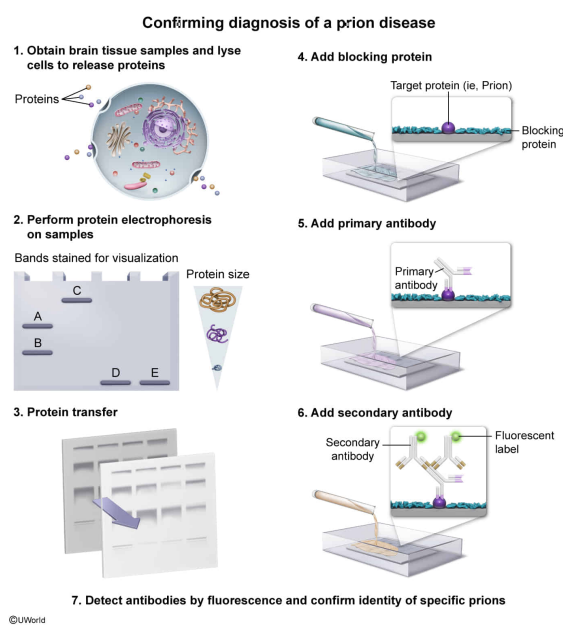
Scrapie is a transmissible neurodegenerative disease in sheep and goats caused by prions. Given this, a diagnosis of scrapie can most likely be confirmed by:

- ☐ A. PCR. [7%]
- ✓ ☒ B. Western blot. [80%]
- ☐ C. DNA sequencing. [5%]
- ☐ D. introduction of specific siRNAs. [6%]

Correct

79% answered correctly

Explanation:



A **prion** is a misfolded protein that acts as an infectious agent by inducing changes in the **secondary structures** of other normal proteins, causing these normal proteins to also become misfolded. These misfolded proteins are less soluble, so they aggregate and can cause disease. Because they are proteins, prions do not contain genetic material (ie, DNA, RNA).

Based on the question, scrapie is a transmissible neurodegenerative disease in sheep and goats known to be caused by prions. Consequently, a diagnosis of scrapie can most likely be confirmed by a procedure that detects specific **proteins** in the brains of these animals. **Western blot**, a method used to detect the presence of specific proteins, is a suitable technique to confirm a diagnosis of scrapie.

To perform this process, the following steps are taken:

1. Obtain brain samples from animals with scrapie and lyse cells to release proteins.
2. Perform gel **electrophoresis** on the protein samples.
3. **Transfer** proteins in the gel to a protein-binding membrane (eg, nitrocellulose), where

they become immobilized.

4. Using protein-rich mixtures, block the portions of the membrane to which proteins were *not* transferred to prevent antibodies (which are also proteins) from **nonspecifically binding** to the membrane in step 6.
5. Incubate membrane with primary antibodies that bind specific prion fragments.
6. Apply secondary antibody with fluorescent label, which binds the primary antibody.
7. **Detect** antibodies by fluorescence or **chemiluminescence** and confirm identity of specific prions.

(Choice A) **Polymerase chain reaction** (PCR) uses thermal cycling to amplify small DNA fragments. Because scrapie is caused by proteins, *not* DNA, PCR is *not* a suitable technique to confirm a scrapie diagnosis.

(Choice C) **DNA sequencing** is used to determine the specific **nucleotide** order of a DNA molecule; therefore, it could *not* be used to detect the proteins responsible for scrapie.

(Choice D) Specific **small interfering RNAs** (siRNAs) introduced into cells can bind complementary nucleotide sequences on target mRNA molecules, causing mRNA degradation or inhibition of **translation**. Scrapie is caused by proteins, *not* RNA, so introduction of siRNAs specific to prions *cannot* confirm a scrapie diagnosis.

Educational objective:

Prions are misfolded proteins that act as infectious agents. Because they are proteins, a western blot can be used to detect the presence of specific prions.

Time spent:0 sec

Subject:
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